Formal Foundations of Reinforcement Learning and Planning

Mathematical Foundations for Agentic AI

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Abstract

These notes develop the mathematical foundations of reinforcement learning (RL) and planning that underpin modern agentic AI systems. Starting from the reward hypothesis and the Bellman optimality principle, we build up to distributional RL, successor features, information-directed sampling, and the options framework for temporal abstraction. Throughout, we emphasize rigorous proofs—the contraction properties of Bellman operators, distributional convergence in the Wasserstein metric, and the Generalised Policy Improvement theorem—while connecting these classical results to contemporary developments in large-scale agentic AI, RLHF, test-time compute, and the "bitter lesson."

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1 The Reward Hypothesis and "Reward is Enough"

1.1 Statement and Formalization

The **reward hypothesis** (Sutton, 2004) asserts:

The Reward Hypothesis

That all of what we mean by goals and purposes can be well thought of as the maximization of the expected value of the cumulative sum of a received scalar signal (called reward).

Silver et al. (2021) strengthened this to the "Reward is Enough" thesis: intelligence and its associated abilities—knowledge, learning, perception, social intelligence, language, generalization—arise as subgoals of reward maximization in a *sufficiently rich environment*.

Definition 1.1 (Sufficiently Rich Environment)**.** *An environment* $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma)$ *is* sufficiently rich *for a capability* $\mathfrak{C}$ *if every policy* π *that achieves near-optimal expected return* $\mathbb{E}_\pi[\sum_{t=0}^{\infty} \gamma^t R_t] \geq V^* - \varepsilon$ *necessarily exhibits capability* $\mathfrak{C}$*. Formally, there exists a predicate* $\Phi_{\mathfrak{C}}$ *on policies such that*

$$V^\pi \geq V^* - \varepsilon \implies \Phi_{\mathfrak{C}}(\pi) = \textit{true}.$$

The key insight is that the environment does the "heavy lifting": in a complex enough world, achieving high reward *requires* developing sophisticated internal representations, planning capabilities, and even social reasoning.

1.2 Connection to AIXI

Hutter's AIXI framework (2004) formalizes the reward-maximization hypothesis in its most general form. The AIXI agent selects actions to maximize expected future reward under a universal prior:

$$a_t^{\text{AIXI}} = \arg\max_{a_t} \sum_{o_t r_t} \cdots \max_{a_m} \sum_{o_m r_m} [r_t + \cdots + r_m] \sum_{\nu \in \mathcal{M}_U} 2^{-K(\nu)} \nu(o_{1:m} r_{1:m} \mid a_{1:m}),$$

where $K(\nu)$ is the Kolmogorov complexity of environment ν, and $\mathcal{M}_U$ is the class of all computable environments. AIXI is Pareto-optimal but incomputable—it establishes reward maximization as *sufficient* in principle, while leaving the question of *efficient* reward maximization to practical RL.

1.3 Counterarguments: Multi-Objective RL

The reward hypothesis faces substantive objections:

(i) **Multi-objective trade-offs.** Real agents often face Pareto-incomparable objectives (e.g., safety vs. performance). While scalarization $R = \sum_i w_i R_i$ can in principle recover any Pareto-optimal policy, the *choice* of weights w itself requires a meta-objective (Roijers et al., 2013).

(ii) **Non-Markovian rewards.** Many natural objectives (e.g., fairness over time) are not expressible as functions of the current state-action pair. Reward machines (Icarte et al., 2018) and temporal logic specifications partially address this.

(iii) **Specification gaming.** Optimizing a proxy reward may diverge from the designer's true intent—the *reward misspecification* problem central to AI alignment.

(iv) **Bounded rationality.** Real agents have computational constraints that prevent exact reward maximization; satisficing or resource-bounded rationality may be more descriptive (Simon, 1955).

Despite these objections, the reward hypothesis remains the dominant paradigm and provides the mathematical scaffolding for RL theory.

2 Bellman Optimality and Dynamic Programming

2.1 MDP Formalism

Definition 2.1 (Markov Decision Process)**.** *A (discounted) MDP is a tuple $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma)$ where:*

- *$\mathcal{S}$ is a (finite) state space, $\mathcal{A}$ is a (finite) action space.*
- *$P : \mathcal{S} \times \mathcal{A} \to \Delta(\mathcal{S})$ is the transition kernel: $P(s' \mid s, a) = \Pr(S_{t+1} = s' \mid S_t = s, A_t = a)$.*
- *$R : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ is the expected reward function (or more generally, $R : \mathcal{S} \times \mathcal{A} \to \Delta(\mathbb{R})$).*
- *$\gamma \in [0, 1)$ is the discount factor.*

Definition 2.2 (Value Functions)**.** *For a stationary policy $\pi : \mathcal{S} \to \Delta(\mathcal{A})$:*

$$V^\pi(s) = \mathbb{E}_\pi\left[\sum_{t=0}^{\infty} \gamma^t R(S_t, A_t) \;\middle|\; S_0 = s\right], \tag{1}$$

$$Q^\pi(s, a) = \mathbb{E}_\pi\left[\sum_{t=0}^{\infty} \gamma^t R(S_t, A_t) \;\middle|\; S_0 = s, A_0 = a\right]. \tag{2}$$

2.2 Bellman Equations

The value functions satisfy the *Bellman consistency equations*:

$$V^\pi(s) = \sum_a \pi(a \mid s)\left[R(s, a) + \gamma \sum_{s'} P(s' \mid s, a)\, V^\pi(s')\right], \tag{3}$$

$$Q^\pi(s, a) = R(s, a) + \gamma \sum_{s'} P(s' \mid s, a) \sum_{a'} \pi(a' \mid s')\, Q^\pi(s', a'). \tag{4}$$

The *Bellman optimality equations* characterize the optimal value:

$$V^*(s) = \max_{a \in \mathcal{A}}\left[R(s, a) + \gamma \sum_{s'} P(s' \mid s, a)\, V^*(s')\right], \tag{5}$$

$$Q^*(s, a) = R(s, a) + \gamma \sum_{s'} P(s' \mid s, a) \max_{a'} Q^*(s', a'). \tag{6}$$

2.3 The Bellman Operator and Contraction

Definition 2.3 (Bellman Optimality Operator)**.** *Define $\mathcal{T} : \mathbb{R}^{|\mathcal{S}|} \to \mathbb{R}^{|\mathcal{S}|}$ by*

$$(\mathcal{T}V)(s) = \max_{a \in \mathcal{A}}\left[R(s, a) + \gamma \sum_{s'} P(s' \mid s, a)\, V(s')\right].$$

Theorem 2.4 (γ-Contraction in Supremum Norm)**.** *The Bellman optimality operator $\mathcal{T}$ is a γ-contraction in the ℓ_∞ norm: for all $V, U \in \mathbb{R}^{|\mathcal{S}|}$,*

$$\|\mathcal{T}V - \mathcal{T}U\|_\infty \le \gamma \|V - U\|_\infty.$$

Proof. For any state $s \in \mathcal{S}$, let $a^* = \arg\max_a [R(s,a) + \gamma \sum_{s'} P(s'|s,a)V(s')]$. Then:

$$\begin{aligned}
(\mathcal{T}V)(s) - (\mathcal{T}U)(s) &= \max_a \big[R(s,a) + \gamma \textstyle\sum_{s'} P(s'|s,a)V(s')\big] - \max_a \big[R(s,a) + \gamma \textstyle\sum_{s'} P(s'|s,a)U(s')\big] \\
&\le R(s,a^*) + \gamma \textstyle\sum_{s'} P(s'|s,a^*)V(s') - R(s,a^*) - \gamma \textstyle\sum_{s'} P(s'|s,a^*)U(s') \\
&= \gamma \textstyle\sum_{s'} P(s'|s,a^*)\big[V(s') - U(s')\big] \\
&\le \gamma \textstyle\sum_{s'} P(s'|s,a^*)\, \|V - U\|_\infty \\
&= \gamma \|V - U\|_\infty.
\end{aligned}$$

By symmetry (swapping V and U), we also obtain $(\mathcal{T}U)(s) - (\mathcal{T}V)(s) \le \gamma\|V - U\|_\infty$. Hence $|(\mathcal{T}V)(s) - (\mathcal{T}U)(s)| \le \gamma\|V - U\|_\infty$ for all s, and taking the supremum over s yields the result. □

Corollary 2.5 (Existence and Uniqueness of V^*)**.** *By the Banach fixed-point theorem, since $(\mathbb{R}^{|\mathcal{S}|}, \|\cdot\|_\infty)$ is a complete metric space and $\mathcal{T}$ is a γ-contraction, there exists a unique $V^* \in \mathbb{R}^{|\mathcal{S}|}$ satisfying $\mathcal{T}V^* = V^*$. Moreover, value iteration $V_{k+1} = \mathcal{T}V_k$ converges geometrically: $\|V_k - V^*\|_\infty \le \gamma^k \|V_0 - V^*\|_\infty$.*

2.4 The Deadly Triad

Sutton and Barto (2018) identified the **deadly triad**: the combination of

1. **Function approximation** (representing V or Q with a parameterized function class),
2. **Bootstrapping** (updating estimates from other estimates, as in TD learning),
3. **Off-policy learning** (learning about a policy different from the behavior policy),

which can cause *divergence* of the learning process. Any two of the three are safe; the combination of all three can be unstable.

Theorem 2.6 (Linear TD Convergence, Tsitsiklis and Van Roy, 1997)**.** *Consider on-policy TD(0) with linear function approximation $\hat{V}(s;\mathbf{w}) = \boldsymbol{\phi}(s)^\top \mathbf{w}$, where $\boldsymbol{\phi}(s) \in \mathbb{R}^d$ is a feature vector. Let $\boldsymbol{\Phi} \in \mathbb{R}^{|\mathcal{S}|\times d}$ be the feature matrix and $\mathbf{D} = \mathrm{diag}(d_\pi(s))$ the diagonal matrix of the stationary distribution. If $\boldsymbol{\Phi}$ has full column rank, then the TD fixed point*

$$\mathbf{w}^* = (\boldsymbol{\Phi}^\top \mathbf{D}(\boldsymbol{\Phi} - \gamma \mathbf{P}_\pi \boldsymbol{\Phi}))^{-1} \boldsymbol{\Phi}^\top \mathbf{D}\, \mathbf{r}_\pi$$

exists and is unique, and TD(0) converges to $\mathbf{w}^*$ with probability 1 under standard stochastic approximation conditions (Robbins–Monro step sizes). The approximation error satisfies:

$$\|\boldsymbol{\Phi}\mathbf{w}^* - V^\pi\|_{\mathbf{D}} \le \frac{1}{\sqrt{1-\gamma^2}} \min_{\mathbf{w}} \|\boldsymbol{\Phi}\mathbf{w} - V^\pi\|_{\mathbf{D}}.$$

3 Distributional Reinforcement Learning

3.1 Return Distributions

Rather than modeling the *expected* return, distributional RL (Bellemare, Dabney, and Munos, 2017) models the full *distribution* of the random return.

Definition 3.1 (Return Distribution)**.** *For a policy π, the random return from state s taking action a is*

$$Z^\pi(s,a) = \sum_{t=0}^{\infty} \gamma^t R(S_t, A_t), \quad S_0 = s,\ A_0 = a,\ A_t \sim \pi(\cdot|S_t) \text{ for } t \ge 1.$$

The value function is recovered as $Q^\pi(s,a) = \mathbb{E}[Z^\pi(s,a)]$.

3.2 Distributional Bellman Equation

The return distribution satisfies a *distributional* Bellman equation:

$$Z^\pi(s,a) \overset{D}{=} R(s,a) + \gamma\, Z^\pi(S', A'), \quad S' \sim P(\cdot \mid s,a),\ A' \sim \pi(\cdot \mid S').$$

Here $\overset{D}{=}$ denotes equality in distribution. Define the **distributional Bellman operator** $\mathcal{T}^\pi$ acting on return-distribution functions $Z : \mathcal{S} \times \mathcal{A} \to \mathcal{P}(\mathbb{R})$:

$$(\mathcal{T}^\pi Z)(s,a) \overset{D}{=} R(s,a) + \gamma\, Z(S', A').$$

3.3 Contraction in Wasserstein Distance

Definition 3.2 (p-Wasserstein Distance)**.** *For probability measures μ, ν on $\mathbb{R}$ with finite p-th moments, the p-Wasserstein distance is:*

$$W_p(\mu,\nu) = \inf_{\text{coupling } (X,Y)} \big(\mathbb{E}[|X-Y|^p]\big)^{1/p},$$

where the infimum is over all joint distributions of (X,Y) with marginals μ and ν. For distributions on $\mathbb{R}$, via the quantile representation: $W_p(\mu,\nu) = \big(\int_0^1 |F_\mu^{-1}(\tau) - F_\nu^{-1}(\tau)|^p\, d\tau\big)^{1/p}$.

We lift the Wasserstein distance to return-distribution functions. For $Z_1, Z_2 : \mathcal{S} \times \mathcal{A} \to \mathcal{P}(\mathbb{R})$:

$$\bar{d}_p(Z_1, Z_2) = \sup_{s,a} W_p\big(Z_1(s,a),\ Z_2(s,a)\big).$$

Theorem 3.3 (Distributional Contraction, Bellemare et al., 2017)**.** *The distributional Bellman operator $\mathcal{T}^\pi$ is a γ-contraction in $\bar{d}_p$ for any $p \geq 1$:*

$$\bar{d}_p(\mathcal{T}^\pi Z_1,\ \mathcal{T}^\pi Z_2) \leq \gamma\, \bar{d}_p(Z_1, Z_2).$$

Proof (coupling argument). Fix $(s,a) \in \mathcal{S} \times \mathcal{A}$. We have

$$(\mathcal{T}^\pi Z_1)(s,a) \overset{D}{=} R(s,a) + \gamma\, Z_1(S', A'),$$
$$(\mathcal{T}^\pi Z_2)(s,a) \overset{D}{=} R(s,a) + \gamma\, Z_2(S', A'),$$

where $S' \sim P(\cdot|s,a)$ and $A' \sim \pi(\cdot|S')$.

We construct a coupling. For each (s', a'), let $(X_{s',a'}, Y_{s',a'})$ be an *optimal coupling* of $Z_1(s',a')$ and $Z_2(s',a')$, i.e.,

$$\mathbb{E}[|X_{s',a'} - Y_{s',a'}|^p] = W_p^p(Z_1(s',a'), Z_2(s',a')).$$

Now couple $(\mathcal{T}^\pi Z_1)(s,a)$ and $(\mathcal{T}^\pi Z_2)(s,a)$ by using the *same* transition (S', A') for both and using the optimal coupling for the downstream returns. This yields random variables $\hat{X} = R(s,a) + \gamma X_{S',A'}$ and $\hat{Y} = R(s,a) + \gamma Y_{S',A'}$ that are valid couplings. Then:

$$\begin{aligned}
W_p^p\big((\mathcal{T}^\pi Z_1)(s,a),\ (\mathcal{T}^\pi Z_2)(s,a)\big) &\leq \mathbb{E}\big[|\hat{X} - \hat{Y}|^p\big] \\
&= \mathbb{E}\big[|\gamma X_{S',A'} - \gamma Y_{S',A'}|^p\big] \\
&= \gamma^p\, \mathbb{E}\big[|X_{S',A'} - Y_{S',A'}|^p\big] \\
&= \gamma^p \sum_{s',a'} P(s'|s,a)\, \pi(a'|s')\, W_p^p(Z_1(s',a'), Z_2(s',a')) \\
&\leq \gamma^p\, \bar{d}_p(Z_1, Z_2)^p.
\end{aligned}$$

Taking p-th roots and the supremum over (s,a):

$$\bar{d}_p(\mathcal{T}^\pi Z_1, \mathcal{T}^\pi Z_2) \leq \gamma\, \bar{d}_p(Z_1, Z_2). \qquad \square$$

Remark 3.4. *By the Banach fixed-point theorem applied to the complete metric space $(\mathcal{Z}, \bar{d}_p)$ (where $\mathcal{Z}$ is the space of return-distribution functions with finite p-th moments), the distributional Bellman equation has a unique fixed point Z^π. Crucially, the distributional Bellman* optimality *operator is **not** a contraction in any $\bar{d}_p$, which introduces additional algorithmic challenges.*

3.4 Algorithmic Instantiations

- **C51** (Bellemare et al., 2017): Represents $Z(s,a)$ as a categorical distribution over a fixed set of $N = 51$ atoms $\{z_1, \ldots, z_N\}$. The distributional Bellman update is projected onto this support.

- **QR-DQN** (Dabney et al., 2018a): Parameterizes the return distribution by N quantile values $\theta_1, \ldots, \theta_N$ at fixed quantile levels $\tau_i = (2i-1)/(2N)$. Minimizes the quantile Huber loss:

$$\sum_{i=1}^{N} \sum_{j=1}^{N'} \rho_{\tau_i}^{\kappa}(\delta_{ij}), \quad \rho_{\tau}^{\kappa}(u) = |\tau - \mathbf{1}_{u<0}| \cdot L_{\kappa}(u),$$

  where L_κ is the Huber loss and $\delta_{ij} = r + \gamma\theta'_j - \theta_i$.

- **IQN** (Dabney et al., 2018b): Implicit Quantile Network—instead of fixed quantile levels, samples $\tau \sim \text{Uniform}(0,1)$ and learns the quantile function $F_Z^{-1}(\tau; s, a)$ directly using a neural network that takes τ as input.

3.5 Neuroscience Connection

Distributional Coding in the Brain

Dabney et al. (2020, *Nature*) demonstrated that dopamine neurons in the mouse brain encode not just reward prediction errors (as in classical TD learning), but a *distribution* of prediction errors. Different neurons appear to encode different quantiles of the return distribution, forming a population code for distributional value. This provides striking biological evidence for the normative optimality of distributional RL: evolution arrived at distributional representations independently.

4 Successor Features and Generalised Policy Improvement

4.1 Successor Representation

The **successor representation** (Dayan, 1993) decomposes the value function by separating *what happens* (dynamics under π) from *what is rewarding* (the reward function).

Assume rewards are linear in state features: $R(s,a) = \boldsymbol{\phi}(s,a)^\top \mathbf{w}$, where $\boldsymbol{\phi} : \mathcal{S} \times \mathcal{A} \to \mathbb{R}^d$ and $\mathbf{w} \in \mathbb{R}^d$.

Definition 4.1 (Successor Features)**.** *The successor features (SFs) under policy π are:*

$$\boldsymbol{\psi}^\pi(s,a) = \mathbb{E}_\pi\left[\sum_{t=0}^{\infty} \gamma^t \boldsymbol{\phi}(S_t, A_t) \,\middle|\, S_0 = s, A_0 = a\right] \in \mathbb{R}^d.$$

The Q-function decomposes as:

$$Q^\pi(s,a) = \boldsymbol{\psi}^\pi(s,a)^\top \mathbf{w}. \tag{7}$$

Critically, $\boldsymbol{\psi}^\pi$ depends only on π and the dynamics P, not on $\mathbf{w}$. Given a library of SFs $\{\boldsymbol{\psi}^{\pi_i}\}_{i=1}^n$ computed for different policies, a new task specified by $\mathbf{w}'$ can be solved *instantly* by evaluating $\boldsymbol{\psi}^{\pi_i}(s,a)^\top \mathbf{w}'$ for each stored policy.

4.2 Generalised Policy Improvement

Theorem 4.2 (Generalised Policy Improvement, Barreto et al., 2017)**.** *Let $\pi_1, \ldots, \pi_n$ be n policies with known successor features $\boldsymbol{\psi}^{\pi_1}, \ldots, \boldsymbol{\psi}^{\pi_n}$, and let $\mathbf{w}$ specify a new task with $R(s,a) = \boldsymbol{\phi}(s,a)^\top \mathbf{w}$. Define the GPI policy:*

$$\pi_{\text{GPI}}(s) = \arg\max_{a \in \mathcal{A}} \max_{i \in \{1,\ldots,n\}} \boldsymbol{\psi}^{\pi_i}(s,a)^\top \mathbf{w}.$$

Then π_{GPI} is at least as good as every π_i on task $\mathbf{w}$:

$$Q^{\pi_{\text{GPI}}}(s,a) \geq \max_i Q^{\pi_i}(s,a) \quad \forall\, s \in \mathcal{S},\ a \in \mathcal{A}.$$

In particular, $V^{\pi_{\text{GPI}}}(s) \geq \max_i V^{\pi_i}(s)$ for all s.

Proof. We proceed by a policy improvement argument. For any $s \in \mathcal{S}$:

$$\begin{aligned} V^{\pi_{\text{GPI}}}(s) &= \mathbb{E}_{\pi_{\text{GPI}}}\left[\sum_{t=0}^{\infty} \gamma^t R(S_t, A_t) \;\middle|\; S_0 = s\right] \\ &= \mathbb{E}_{\pi_{\text{GPI}}}[R(s, A_0) + \gamma V^{\pi_{\text{GPI}}}(S_1) \mid S_0 = s]\,. \end{aligned}$$

By construction, π_{GPI} selects $A_0 = \arg\max_a \max_i \boldsymbol{\psi}^{\pi_i}(s,a)^\top \mathbf{w}$. Thus:

$$Q^{\pi_{\text{GPI}}}(s,a) = R(s,a) + \gamma \mathbb{E}[V^{\pi_{\text{GPI}}}(S_1) \mid S_0 = s, A_0 = a].$$

We prove $Q^{\pi_{\text{GPI}}}(s,a) \geq \max_i Q^{\pi_i}(s,a)$ by induction on the horizon. For the base case ($\gamma = 0$), it is immediate since $Q^{\pi_i}(s,a) = R(s,a)$ for all i.

For the inductive step, assume $V^{\pi_{\text{GPI}}}(s') \geq \max_i V^{\pi_i}(s')$ for all s' reachable in k steps. Then for state s at step $k+1$:

$$\begin{aligned} Q^{\pi_{\text{GPI}}}(s,a) &= R(s,a) + \gamma \sum_{s'} P(s'|s,a)\, V^{\pi_{\text{GPI}}}(s') \\ &\geq R(s,a) + \gamma \sum_{s'} P(s'|s,a) \max_i V^{\pi_i}(s') \\ &\geq R(s,a) + \gamma \sum_{s'} P(s'|s,a)\, V^{\pi_i}(s') \quad \text{for all } i \\ &= Q^{\pi_i}(s,a) \quad \text{for all } i. \end{aligned}$$

Taking the max over i gives the result. The argument extends to the infinite-horizon discounted case via the contraction mapping argument: let Q_{GPI} be the fixed point of the operator $(\mathcal{T}_{\text{GPI}} Q)(s,a) = R(s,a) + \gamma \sum_{s'} P(s'|s,a) \max_a \max_i Q_i(s',a')$, where we can show this operator dominates each $\mathcal{T}^{\pi_i}$ pointwise. □

4.3 Zero-Shot Transfer

The GPI theorem enables **zero-shot transfer**: given successor features from training tasks $\mathbf{w}_1, \ldots, \mathbf{w}_n$, performance on a new task $\mathbf{w}$ is bounded by:

$$V^*(s;\mathbf{w}) - V^{\pi_{\text{GPI}}}(s;\mathbf{w}) \leq \frac{2\gamma}{1-\gamma} \min_i \|\boldsymbol{\psi}^{\pi_i} - \boldsymbol{\psi}^*(s;\mathbf{w})\|_\infty \|\mathbf{w}\|_1.$$

If the training tasks span the reward space well, this gap can be small even for novel tasks.

5 Information-Directed Sampling

5.1 The Exploration–Exploitation Dilemma

The fundamental challenge in sequential decision-making is balancing *exploitation* (acting on current knowledge) with *exploration* (gathering information to improve future decisions). Classical approaches include:

- **UCB (Upper Confidence Bound)**: Act optimistically—select actions whose confidence-upper-bound on value is highest. Achieves $O(\sqrt{KT \log T})$ regret for K-armed bandits (Auer et al., 2002).
- **Thompson Sampling**: Sample a model from the posterior and act greedily with respect to the sample. Achieves $O(\sqrt{KT \log K})$ Bayesian regret (Russo and Van Roy, 2014).

Both can be suboptimal in structured problems. **Information-Directed Sampling** (IDS), due to Russo and Van Roy (2014, 2018), provides a principled alternative.

5.2 The Information Ratio

Definition 5.1 (Information Ratio)**.** *Let π_t be a (possibly randomized) action-selection rule at time t. Given posterior P_t over the environment, define:*

- ***Expected instantaneous regret:*** *$\Delta_t(\pi_t) = \mathbb{E}_{A_t \sim \pi_t}[\mathbb{E}_{P_t}[R(A^*) - R(A_t)]]$, where A^* is the optimal action under the true model.*
- ***Information gain:*** *$g_t(\pi_t) = I_{P_t}(\theta; (A_t, O_t) \mid A_t \sim \pi_t)$, the mutual information between the unknown parameter θ and the observation O_t under action A_t.*

*The **information ratio** is:*

$$\Gamma_t(\pi_t) = \frac{\Delta_t(\pi_t)^2}{g_t(\pi_t)}. \tag{8}$$

Definition 5.2 (Information-Directed Sampling)**.** *The IDS policy minimizes the information ratio:*

$$\pi_t^{\mathrm{IDS}} = \arg\min_{\pi_t \in \Delta(\mathcal{A})} \Gamma_t(\pi_t) = \arg\min_{\pi_t} \frac{\Delta_t(\pi_t)^2}{g_t(\pi_t)}.$$

This trades off regret and information optimally: IDS takes actions that yield the best ratio of "regret squared per bit of information gained."

5.3 Regret Bound

Theorem 5.3 (IDS Regret Bound, Russo and Van Roy, 2014)**.** *If $\Gamma_t(\pi_t^{\mathrm{IDS}}) \le \bar{\Gamma}$ for all t (uniformly bounded information ratio), and $H(\theta) < \infty$ is the entropy of the prior, then the Bayesian regret of IDS satisfies:*

$$\mathrm{Regret}(T) = \mathbb{E}\left[\sum_{t=1}^{T} \Delta_t\right] \le \sqrt{\bar{\Gamma} \cdot H(\theta) \cdot T}.$$

Proof sketch. By the definition of the information ratio: $\Delta_t^2 \le \bar{\Gamma} \cdot g_t$ for all t. Summing over t and applying Jensen's inequality:

$$\left(\frac{1}{T}\sum_{t=1}^{T} \Delta_t\right)^2 \le \frac{1}{T}\sum_{t=1}^{T} \Delta_t^2 \le \frac{\bar{\Gamma}}{T}\sum_{t=1}^{T} g_t.$$

The key insight is the *information-theoretic chain rule*: the total information gained about θ cannot exceed the prior entropy, $\sum_{t=1}^{T} g_t \le H(\theta)$. This is because $I(\theta; O_{1:T}) \le H(\theta)$, and the

chain rule for mutual information gives $I(\theta; O_{1:T}) = \sum_t I(\theta; O_t \mid O_{1:t-1}) = \sum_t g_t$ (adapting for the adaptive action selection). Hence:

$$\text{Regret}(T) = \sum_{t=1}^{T} \Delta_t \leq \sqrt{T \sum_{t=1}^{T} \Delta_t^2} \leq \sqrt{T \cdot \bar{\Gamma} \cdot H(\theta)}. \qquad \square$$

Remark 5.4. *For K-armed Bernoulli bandits, IDS achieves $\bar{\Gamma} \leq K/2$, yielding $O(\sqrt{KT \log K})$ Bayesian regret (matching Thompson sampling). For linear bandits in d dimensions, $\bar{\Gamma} \leq d/2$, giving $O(d\sqrt{T})$ regret that can* improve *upon both UCB and Thompson sampling in structured problems.*

5.4 Comparison and Connection to Active Inference

Method	**Principle**	**Regret (K-armed)**	**Structured?**
UCB	Optimism	$O(\sqrt{KT \log T})$	Moderate
Thompson Sampling	Posterior sampling	$O(\sqrt{KT \log K})$	Good
IDS	Min info ratio	$O(\sqrt{KT \log K})$	Excellent

IDS is closely related to the **active inference** framework from neuroscience (Friston, 2010), which postulates that biological agents minimize *expected free energy*—a quantity that decomposes into an exploitation term (expected reward or pragmatic value) and an exploration term (expected information gain or epistemic value). IDS can be viewed as a decision-theoretic formalization of this decomposition.

6 The Options Framework

6.1 Temporal Abstraction

Primitive actions operate at the finest time scale. Many tasks are better described at multiple temporal scales: "navigate to the kitchen" is a temporally extended action composed of many primitive steps. The **options framework** (Sutton, Precup, and Singh, 1999) formalizes this via *temporal abstraction.*

Definition 6.1 (Option)**.** *An option $o = (\mathcal{I}_o, \pi_o, \beta_o)$ consists of:*

- *$\mathcal{I}_o \subseteq \mathcal{S}$: the **initiation set** (states where o can begin),*
- *$\pi_o : \mathcal{S} \to \Delta(\mathcal{A})$: the **intra-option policy**,*
- *$\beta_o : \mathcal{S} \to [0, 1]$: the **termination function** ($\beta_o(s)$ is the probability of terminating in state s).*

Primitive actions are options with $\mathcal{I}_o = \mathcal{S}$, $\pi_o(\cdot|s) = \mathbf{1}_a$, and $\beta_o \equiv 1$ (immediate termination).

6.2 Semi-MDP Formulation

When executing options, the time between decisions is variable. This gives rise to a **semi-MDP** (SMDP) over options. For a policy-over-options $\mu : \mathcal{S} \to \Delta(\mathcal{O})$ (where $\mathcal{O}$ is the set of available options), the **option-value function** satisfies the SMDP Bellman equation:

$$Q^{\mu}_{\mathcal{O}}(s, o) = \mathbb{E}\left[\sum_{k=0}^{\tau_o - 1} \gamma^k R_{t+k} + \gamma^{\tau_o} \sum_{o'} \mu(o' | S_{t+\tau_o})\, Q^{\mu}_{\mathcal{O}}(S_{t+\tau_o}, o') \;\middle|\; S_t = s, O_t = o \right], \tag{9}$$

where τ_o is the (random) duration of option o.

With the *call-and-return* option execution model, this can be written more compactly using the **option-conditional transition model**:

$$P_o(s', k \mid s) = \Pr(S_{t+k} = s', \tau_o = k \mid S_t = s),$$

and the expected discounted reward during execution:

$$r_o(s) = \mathbb{E}\left[\sum_{k=0}^{\tau_o - 1} \gamma^k R_{t+k} \;\middle|\; S_t = s\right].$$

6.3 Eigenoptions and Option Discovery

A key challenge is *discovering* useful options automatically. **Eigenoptions** (Machado et al., 2017) uses the graph Laplacian of the state-transition graph: eigenvectors of the Laplacian identify bottleneck states and natural sub-goals.

Given the graph Laplacian $\mathbf{L} = \mathbf{D}_G - \mathbf{A}$ (where $\mathbf{A}$ is the adjacency matrix and $\mathbf{D}_G$ the degree matrix), the Fiedler vector (second-smallest eigenvector of $\mathbf{L}$) partitions the state space into two weakly connected components. Options that navigate between these components correspond to traversing bottleneck states and are useful for exploration.

More generally, the k-th eigenvector $\mathbf{e}_k$ of $\mathbf{L}$ defines an intrinsic reward $r_k(s, s') = \mathbf{e}_k(s') - \mathbf{e}_k(s)$, and the corresponding eigenoption is the policy maximizing cumulative r_k.

6.4 Option-Critic Architecture

The **Option-Critic** (Bacon, Harb, and Precup, 2017) learns options end-to-end using the policy gradient theorem extended to options:

$$\frac{\partial V_{\mathcal{O}}(s)}{\partial \theta_{\pi_o}} \propto \sum_{s,o} \mu_{\mathcal{O}}(s, o \mid s_0) \sum_a \frac{\partial \pi_o(a|s)}{\partial \theta_{\pi_o}} Q_U(s, o, a), \tag{10}$$

$$\frac{\partial V_{\mathcal{O}}(s)}{\partial \theta_{\beta_o}} \propto -\sum_{s',o} \mu_{\mathcal{O}}(s', o \mid s_0) \frac{\partial \beta_o(s')}{\partial \theta_{\beta_o}} A_{\mathcal{O}}(s', o), \tag{11}$$

where $Q_U(s, o, a)$ is the value of executing action a within option o at state s, and $A_{\mathcal{O}}(s', o) = Q_{\mathcal{O}}(s', o) - V_{\mathcal{O}}(s')$ is the advantage of continuing option o.

6.5 Tool Use as Options

In agentic AI systems, **tool use** maps naturally to the options framework:

- A *tool* (web search, code execution, calculator) is an option with its own internal policy (the tool's execution logic).
- The *initiation set* corresponds to contexts where the tool is applicable.
- The *termination condition* is when the tool returns a result.
- The agent's *policy over options* is the decision of which tool to invoke.

This abstraction allows treating tool-augmented agents within the standard RL/planning framework (see Section 8).

7 The Bitter Lesson and Scaling Laws

7.1 The Bitter Lesson

Sutton's "Bitter Lesson" (2019) observes that, over the history of AI research, methods that *leverage computation*—particularly search and learning—have consistently outperformed methods that *encode human knowledge*. This lesson is "bitter" because researchers repeatedly invest

effort in domain-specific engineering, only to be surpassed by general methods that scale with compute.

The key examples span decades: chess (Deep Blue to AlphaZero), Go (handcrafted features to AlphaGo/AlphaZero), speech recognition (HMMs to deep learning), computer vision (SIFT/HOG to CNNs), and NLP (parse trees to Transformers).

7.2 Scaling Laws for RL

Scaling laws in supervised learning (Kaplan et al., 2020; Hoffmann et al., 2022) describe power-law relationships between loss and compute/data/parameters. Analogous scaling phenomena appear in RL:

1. **Sample complexity**: For tabular MDPs, PAC-MDP bounds scale as $\tilde{O}(|\mathcal{S}|^2|\mathcal{A}|/\varepsilon^2)$ (Strehl et al., 2009). With function approximation, this depends on the effective dimension of the function class.
2. **Compute scaling in games**: AlphaZero's Elo rating scales log-linearly with training compute. MuZero exhibits similar scaling.
3. **Test-time compute scaling**: In reasoning tasks, increasing search budget at inference time (e.g., more MCTS rollouts) yields predictable performance improvements following power-law curves.

7.3 The Era of Experience

Silver, Sutton, and collaborators (2025) argue for an "Era of Experience" in which AI agents learn primarily from *self-generated experience* rather than human-curated data. Key claims include:

- The data bottleneck for LLMs (limited human text) can be overcome by agents generating their own training signal through interaction.
- The combination of world models, planning, and self-play enables superhuman performance without human demonstrations.
- This extends the bitter lesson: not only computation, but *experience* should be scaled, and the most general-purpose approach is to let agents learn from their own interaction data.

8 Connections to Modern Agentic AI

The RL foundations developed above connect directly to the architecture and training of modern agentic AI systems.

8.1 RLHF and RLAIF

Reinforcement Learning from Human Feedback (RLHF; Christiano et al., 2017; Ouyang et al., 2022) trains a reward model from human preference comparisons, then optimizes a language model policy against this reward using RL (typically PPO). The objective is:

$$\max_{\pi} \mathbb{E}_{x\sim\mathcal{D},\, y\sim\pi(\cdot|x)}\big[r_\phi(x,y)\big] - \beta\ \mathrm{KL}\big[\pi(\cdot|x)\,\|\,\pi_{\mathrm{ref}}(\cdot|x)\big],$$

where r_ϕ is the learned reward model and the KL penalty prevents collapse away from the reference policy π_{ref}.

RLAIF (Bai et al., 2022) replaces human preferences with AI-generated preferences, connecting to the "Era of Experience" thesis: the agent generates its own training signal.

The closed-form solution of the KL-constrained problem is:

$$\pi^*(y|x) = \frac{1}{Z(x)}\pi_{\mathrm{ref}}(y|x)\exp\left(\frac{r_\phi(x,y)}{\beta}\right),$$

which motivates Direct Preference Optimization (DPO; Rafailov et al., 2023).

8.2 In-Context Learning as Implicit RL

Transformer-based models exhibit *in-context learning*: they adapt their behavior based on examples in the prompt without weight updates. This can be viewed as *implicit* RL or Bayesian inference:

- The prompt context is the "trajectory" so far.
- The model's next-token prediction implements a policy conditioned on history.
- Transformers trained on diverse tasks can implement in-context algorithms that approximate Bayes-optimal policies (Müller et al., 2022; Garg et al., 2022).
- Algorithm distillation (Laskin et al., 2022) explicitly trains Transformers on RL learning histories, enabling in-context RL.

8.3 Test-Time Compute as MCTS

Recent work on "thinking" models (e.g., chain-of-thought, tree-of-thought) allocates additional computation at inference time to solve harder problems. This is directly analogous to MCTS:

- **Chain-of-thought**: A single rollout (analogous to a single MCTS simulation).
- **Tree-of-thought**: Multiple branching rollouts with backtracking (explicit tree search).
- **Best-of-N sampling**: Sample N complete solutions and select the best (analogous to root parallelization in MCTS).
- The value function in MCTS corresponds to the model's self-evaluation capability (process reward models).

8.4 Multi-Step Tasks as POMDPs

Agentic tasks (web browsing, coding, research) are naturally modeled as **POMDPs**: the agent observes partial information (screen content, file contents, API responses) and must maintain beliefs about the hidden state (codebase structure, user intent, task progress). The agent's "scratchpad" or "memory" serves as a belief state, and tool use provides observations that reduce uncertainty.

8.5 Multi-Agent RL and LLM Agents

Littman's framework of **Markov games** (1994) extends MDPs to multiple agents:

Definition 8.1 (Markov Game)**.** *A Markov game for n agents is $(\mathcal{S}, \mathcal{A}_1, \ldots, \mathcal{A}_n, P, R_1, \ldots, R_n, \gamma)$, where $P : \mathcal{S} \times \mathcal{A}_1 \times \cdots \times \mathcal{A}_n \to \Delta(\mathcal{S})$ and each R_i is agent i's reward function.*

Multi-agent LLM systems (e.g., debate, negotiation, collaborative coding) are instances of Markov games. The solution concept shifts from optimality to *equilibria* (Nash, correlated, or coarse-correlated equilibria), and convergence guarantees become substantially more challenging.

9 Open Problems

Open Problem 1 (Non-Stationarity in RLHF)**.** *Human preferences evolve over time, and reward models trained on past preferences may become stale. How should RL agents adapt to non-stationary reward functions while maintaining alignment guarantees? Connections to restless bandits (Whittle, 1988) and non-stationary MDPs remain underexplored.*

Open Problem 2 (Sample Complexity of Distributional RL)**.** *While distributional RL empirically outperforms expectation-based methods, the theoretical sample complexity advantages are not well understood. Under what conditions does learning the full return distribution provably improve policy optimization, beyond the benefits of auxiliary representation learning?*

Open Problem 3 (Scalable Option Discovery). *Current option discovery methods (eigenoptions, interest functions, information-theoretic objectives) struggle to scale to high-dimensional, continuous state spaces encountered in real-world agentic tasks. Can foundation models serve as priors for option discovery, identifying meaningful temporal abstractions from language descriptions?*

Open Problem 4 (Theoretical Foundations of Test-Time Compute Scaling). *Empirical scaling laws for test-time compute (more "thinking" = better performance) lack rigorous theoretical grounding. What are the computational complexity-theoretic characterizations of problems where test-time search provably helps? How does the value of additional computation decay, and what is the optimal compute allocation between training and inference?*

Open Problem 5 (Credit Assignment in Long-Horizon Agentic Tasks). *Agentic tasks (multi-step coding, research, web navigation) involve hundreds or thousands of steps with sparse, delayed rewards. The credit assignment problem—which actions were responsible for success or failure—is exacerbated at this scale. Existing methods (eligibility traces, hindsight relabeling, attention-based credit) do not scale gracefully. Can successor features or distributional methods provide better credit assignment in these regimes?*

Open Problem 6 (Safe Exploration in Open-Ended Environments). *Real-world agentic AI systems must explore safely—avoiding irreversible actions (deleting files, sending emails, financial transactions) while still learning effectively. The constrained MDP framework provides some tools, but the constraint set itself is often unknown and must be learned. How can IDS-style information-theoretic exploration be combined with safety constraints?*

Open Problem 7 (Bridging Language and Formal Reward Specification). *Natural language instructions are the primary interface for agentic AI, yet translating them into formal reward functions remains error-prone. Reward machines (Icarte et al., 2018) and linear temporal logic offer structured approaches, but grounding language to these formalisms at scale is unsolved. Can LLMs themselves serve as reward specification languages, and if so, what are the alignment implications?*

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